Scientific Reports

https://doi.org/10.1038/s41598-026-52116-3

Article in Press

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Received: 3 March 2026

Accepted: 4 May 2026

Published online: 11 May 2026

Cite this article as: Thang V., Pham P.N.L., Vuong T.T.T. *et al.* Sustainable synthesis of Ni-doped Co_3O_4/graphene from recycled cobalt for efficient oxygen evolution reaction. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-52116-3

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Sustainable Synthesis of Ni-doped Co_3O_4/Graphene from Recycled Cobalt for Efficient Oxygen Evolution Reaction

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Abstract

Developing sustainable and cost-effective electrocatalysts for the oxygen evolution reaction (OER) is essential for efficient water splitting. Herein, Ni-doped Co_3O_4 supported on graphene (Ni-Co_3O_4@G) is synthesized using cobalt recovered from spent lithium-ion batteries via a hydrothermal–calcination approach. The composite exhibits a hierarchical porous structure with uniformly dispersed nanoparticles on conductive graphene sheets, providing abundant active sites and enhanced charge transfer. XPS analysis confirms mixed valence states (Co^{2+}/Co^{3+} and Ni^{2+}/Ni^{3+}) and rich oxygen defects, contributing to improved catalytic activity.

Electrochemical results in alkaline media show that Ni-Co_3O_4@G achieves superior OER performance, delivering a low overpotential (1.66 V vs. RHE at 10 mA cm^{-2}), a small Tafel slope (107.52 mV dec^{-1}), and excellent stability over 3000 cycles. This work demonstrates a green strategy for converting recycled cobalt into efficient OER electrocatalysts for sustainable hydrogen production.

Keywords: Ni-doped Co_3O_4, Recycled cobalt, Graphene composite Sustainable synthesis, Water splitting, Low overpotential.

1. Introduction

The escalating global energy demand necessitates the development of sustainable and efficient energy conversion and storage technologies[1]. Among these, electrochemical water splitting has emerged as a promising strategy for generating clean hydrogen fuel, a critical component for mitigating climate change and achieving energy independence[2–4]. The oxygen evolution reaction (OER), however, remains a kinetic bottleneck in overall water splitting due to its sluggish four-electron transfer process and high overpotential requirement, significantly hindering the efficiency of these systems[5,6]. Consequently, the development of highly active, stable, and cost-effective electrocatalysts for OER is paramount.

Noble metal-based catalysts, such as RuO_2 and IrO_2, currently represent the state-of-the-art for OER, exhibiting excellent activity and stability[7]. However, their scarcity, high cost, and limited availability impede their widespread application[8]. This has driven extensive research into developing non-noble metal alternatives, among which transition metal oxides—particularly cobalt-based materials—have garnered considerable attention as efficient OER electrocatalysts in alkaline media due to their intrinsic catalytic activity, relative abundance, and tunable properties[9–13]. Despite those advantages, the utilization of commercial Co-based precursors certainly rises the synthetic cost of the catalysts, as Co is one of the critical elements in Li-ion batteries' production. Instead, recovered cobalt from spent lithium-ion batteries can act as a valuable alternative, not only reduces dependence on primary resources but also contributes to a more sustainable materials lifecycle. The use of recycled cobalt further enables a closed-loop approach, in which valuable metals are reintroduced into functional materials, thereby minimizing waste generation and supporting the development of circular energy systems.

To further enhance the electrocatalytic performance of Co_3O_4, several strategies have been explored, including doping with other transition metals and incorporating carbon-based supports. Doping Co_3O_4 with nickel (Ni) has proven effective in improving catalytic activity. Studies have shown that Ni dopant can remarkably enhance the activity for both OER and hydrogen evolution reaction (HER) of the catalysts in alkaline environments[14–16]. The incorporation of Ni can modify the electronic structure of Co_3O_4, leading to the formation of abundant high-valence Ni^{3+} and Co^{3+} species on the surface, which are crucial for enhanced OER activity[17] . Moreover, the synergistic effect between metallic nickel and cobalt oxides with nitrogen-doped carbon nanospheres has been demonstrated to yield highly efficient OER electrocatalysts[18].

The integration of carbonaceous materials, such as graphene sheet (G), with metal oxides offers additional benefits by increasing surface area, improving electrical conductivity, and facilitating charge transfer kinetics[19,20]. Carbon materials, for instance, can provide a robust framework that stabilizes metal oxide nanoparticles, preventing agglomeration during OER processes and offering abundant active sites[21]. The unique structure of NiO/ Co_3O_4 nanoparticles decorated on nitrogen-doped carbon (NiO/Co_3O_4@NC) promotes strong interactions between the carbon support and metal oxides, contributing to superior OER performance[22].

While these approaches have been widely studied individually, their integration into a single, sustainable synthesis pathway remains underexplored. Herein, we present a unified strategy that combines cobalt recovery from spent lithium-ion batteries, Ni doping, and graphene hybridization within a green synthesis framework. This integrated approach not only streamlines material processing but also provides new insights into how recycled cobalt sources can have a “second life”. To achieve this, Ni-doped Co_3O_4/graphene (Ni–Co_3O_4/G) was synthesized from recycled cobalt and systematically characterized to evaluate its performance as an OER electrocatalyst. By carefully controlling the doping concentration of nickel and integrating graphene, we anticipate achieving a synergistic effect that enhances both the intrinsic activity of Co_3O_4 and the overall electrochemical performance. This research endeavors to provide valuable insights into designing eco-friendly and high-performance OER catalysts, thereby advancing the practical implementation of electrochemical water splitting for sustainable oxygen production.

2. Experimental Section

2.1 Chemicals

Cobalt metal was recovered from spent lithium-ion batteries. Nitric acid (HNO_3, 65%), $Ni(NO_3)_2.6H_2O$, urea ($CO(NH_2)_2$), ammonia (NH_3, 25%), Graphene sheet (G) and potassium hydroxide (KOH) were purchased from Sigma-Aldrich. Cobalt nitrate hexahydrate ($Co(NO_3)_2.6H_2O$, 99%) was produced from Macklin. Deionized water and ethanol were used for washing.

2.2 Preparation of Recycled Cobalt via DES Leaching and Electrochemical Recovery

The electrolyte was first prepared by selectively leaching cobalt from spent Ni–MH cathode powder using a reline deep eutectic solvent (DES). The reline DES was synthesized by mixing choline chloride (99%, Sigma-Aldrich) and urea (98.6%, Sigma-Aldrich) at a molar ratio of 1:2 at 100 °C until a homogeneous transparent liquid was formed, followed by horn ultrasonication for 80 min. The combination of DES leaching and high-energy horn sonication promotes efficient coordination between Co species and the chloride/urea hydrogen-bond network, thereby enhancing the dissolution kinetics and enabling the preferential extraction of cobalt from the complex Ni–MH cathode matrix, as described in our previous work[23]. Under these conditions, cobalt was selectively dissolved into the DES phase, while most insoluble components remained undissolved and were removed by filtration before electrolysis. The resulting Co-containing DES solution was directly used as the electrolyte for subsequent electrodeposition.

Electrochemical recovery of cobalt was then carried out via galvanostatic electrolysis in a conventional three-electrode configuration. A copper plate (Cu) served as the working electrode, a graphite bar was employed as the counter electrode, and a silver wire (Ag) functioned as the reference electrode. Before deposition, the Cu substrate was sequentially cleaned with ethanol and deionized (DI) water, then dried under ambient conditions. Galvanostatic electrolysis was conducted at a constant current of −200 mA for 24 h at room temperature without agitation unless otherwise specified. After electrolysis, a loosely adherent Co film was observed on the Cu surface with partial spontaneous delamination. The deposited Co was collected by ultrasonically treating the Cu plate in DI water for 30 min to detach the electroformed material fully. The recovered solid was washed twice with ethanol and DI water, respectively, and subsequently dried at 60 °C to obtain the final recycled cobalt product.

2.3 Synthesis of Ni-doped Co_3O_4/G

First, 0.026 g of recycled cobalt metal was dissolved in 10 mL of HNO3 (1:1 with deionized water) at 40–50°C under stirring for 30 minutes in a fume hood, yielding a $Co(NO_3)_2$ solution. The solution was filtered using Whatman No. 1 filter paper and evaporated at 80°C to obtain $Co(NO_3)_2$ powder. Next, 0.0052 g of graphene was dispersed in 50 mL of deionized water by ultrasonication for 30 minutes. A reaction mixture was prepared by dissolving the $Co(NO_3)_2$ powder, 0.016 g of $Ni(NO_3)_2.6H_2O$, 0.016 g of urea, and 0.052 mL of NH3 in 50 mL of deionized water, followed by the addition of the graphene dispersion. The mixture (total volume ~100 mL) was transferred to a 200 mL Teflon-lined autoclave and heated at 180°C for 12 hours. The product was centrifuged at 6000 rpm for 8 minutes, washed three times with deionized water and ethanol, and dried at 60°C. Finally, the dried product was calcined at 400°C for 3 hours (heating rate: 5°C min^{-1}) to obtain Ni-doped Co_3O_4/G.

For comparison, Ni–Co_3O_4@G was also synthesized using commercial cobalt precursors following a similar procedure. Specifically, $Co(NO_3)_2·6H_2O$ was used directly instead of the recycled cobalt source, while all other synthesis conditions were kept identical.

2.4 Materials characterizations

The materials that were synthesized were examined by using a variety of techniques. Powder X-ray diffraction of all the synthetic samples were conducted on Rigaku Smartlab diffractometer with a Cu Kα radiation resource (λ = 1.5418 Å) at the step size of 0.02 ° in the scanning range of 10−70 °. Raman spectroscopy (Horiba), Fourier Transform Infrared Spectroscopy (FTIR), scanning electron microscopy (SEM, Hitachi SU8000). High-resolution transmission electron microscopy (HR-TEM, JEM2100). A Thermo mono model spectrometer was used to get XPS data.

2.5 Electrochemical Measurements

Electrochemical tests were conducted using a CS 310 electrochemical workstation (Corrtest, China) with a three-electrode system in 1 M KOH. A glassy carbon electrode (GCE, 3 mm diameter) coated with the catalyst served as the working electrode, Ag/AgCl (3 M KCl) as the reference electrode, and a Pt wire as the counter electrode. To prepare the working electrode, 5 mg of Ni-doped Co3O4/G was dispersed in 250 µL of ethanol, 750 µL of deionized water, and 20 µL of Nafion (5%) via ultrasonication for 30 minutes. Then, 6 µL of the ink was drop-cast onto the GCE and dried at room temperature. Linear sweep voltammetry (LSV) was conducted at a scan rate of 5 mV s-1, while cyclic voltammetry (CV) was performed at scan rates of 100, 50, 20, 10, and 5 mV s-1. Electrochemical impedance spectroscopy (EIS) measurements were carried out over a frequency range from 0.1 Hz to 100 kHz. All potentials were converted to the reversible hydrogen electrode (RHE) scale using the equation: $E_{(RHE)} = E_{Ag/AgCl} + 0.059\ pH + E^{o}_{Ag/AgCl}$.

3. Results and Discussion

3.1 Material Characterization

The crystallographic information and vibrational properties of the synthesized materials herein were respectively investigated by X-ray diffraction and Raman spectroscopy and are presented in Figure 1. As shown in the XRD patterns (Figure 1a), all the diffraction peaks of Co_3O_4 and Ni-doped Co_3O_4 are well indexed to $Fd\bar{3}m$ space group (PDF No. 00-009-0418) with no extra reflections indicating a high purity achieved. The diffractogram of Ni-doped Co_3O_4@G exhibits an additional low-intensity reflection at 26.5 °, which can be assigned to the small amount of graphene incorporated into the material during the synthesis. The XRD pattern of graphene sheet can be found in the Supporting information (Figure S1a). Noticeably, the reflections corresponding to Co_3O_4 structure show so shift regardless of the modifications, yet the intensity of all the peaks declined upon Ni-doping (Figure S1b). This indicates that the cubic structure, on one hand, is stable throughout the synthesis; on the other hand, loses its crystallinity upon the substitution.

Raman spectroscopy plays as a complementary for X-ray diffraction to confirm our successful synthesis. Signature vibrations of Co_3O_4 at 473 cm^{-1}, 519 cm^{-1}, 610 cm^{-1}, and 683 cm^{-1} are observed in the spectrum of our as-prepared Co_3O_4 (Figure 1b), which is in a God agreement with both Co_3O_4 bulk[24] and nanoparticles[25]. Despite that the partial substitution of Co by Ni has minor impact on the global structure, the intensity of the aforementioned signatures decreases, further confirm the lower crystallinity of the materials.

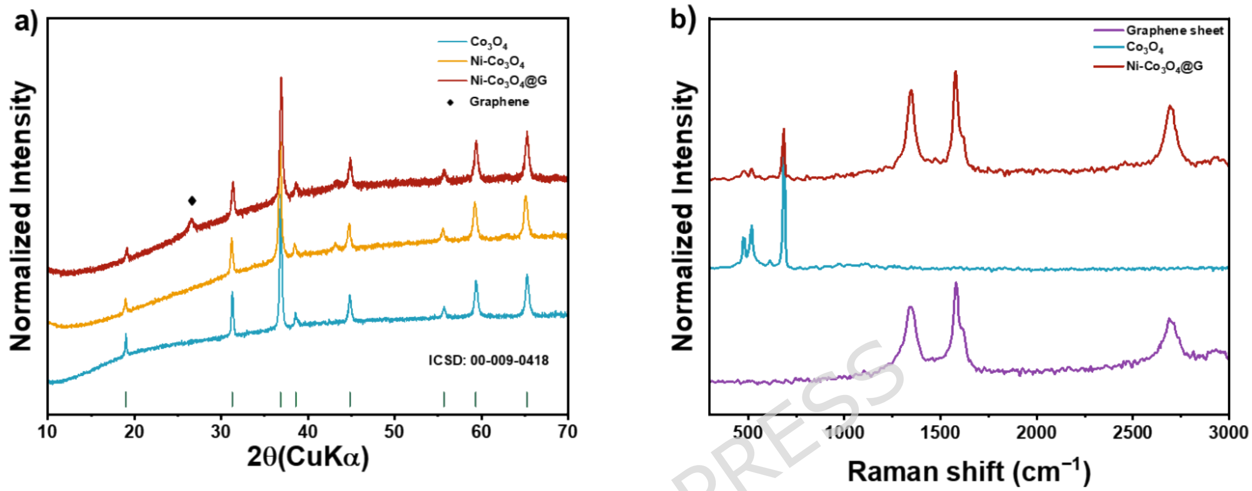

Figure 1. (a) X-ray diffractograms of Co_3O_4, Ni-doped Co_3O_4 and Ni-doped Co_3O_4@G with the Bragg positions of Co_3O_4; (b) Raman spectra of graphene sheet, Co_3O_4, Ni-doped Co_3O_4 and Ni-doped Co_3O_4@G.

Field-emission scanning electron microscopy (FE-SEM) analysis reveals the morphological characteristics of the Ni-doped Co_3O_4@G composite. At low to moderate magnifications (Fig. 2a and b), the material exhibits a highly porous, three-dimensional hierarchical architecture, where thin graphene nanosheets serve as a supportive substrate uniformly decorated with dense aggregates of Ni-doped Co_3O_4 nanoparticles. Large, flake-like graphene sheets (extending up to micrometer scales) are interspersed with clusters of nanoparticles, creating abundant voids and open spaces that facilitate electrolyte accessibility[26]. Higher-magnification images (Fig. 2c and d) clearly show that the Ni-doped Co_3O_4 nanoparticles are predominantly near-spherical in shape, with a relatively uniform size distribution in the range of 40 nm[27]. These primary nanoparticles assemble into secondary hierarchical clusters resembling flower-like or cauliflower-like structures, featuring rough, textured surfaces that significantly enhance the specific surface area[26].

The incorporation of Ni doping combined with the graphene support effectively suppresses particle aggregation and grain growth, resulting in well-dispersed, nanoscale features compared to undoped Co_3O_4 systems[28]. This optimized morphology, characterized by high porosity, hierarchical organization, and excellent dispersion on the conductive graphene matrix, is highly advantageous for electrochemical applications.

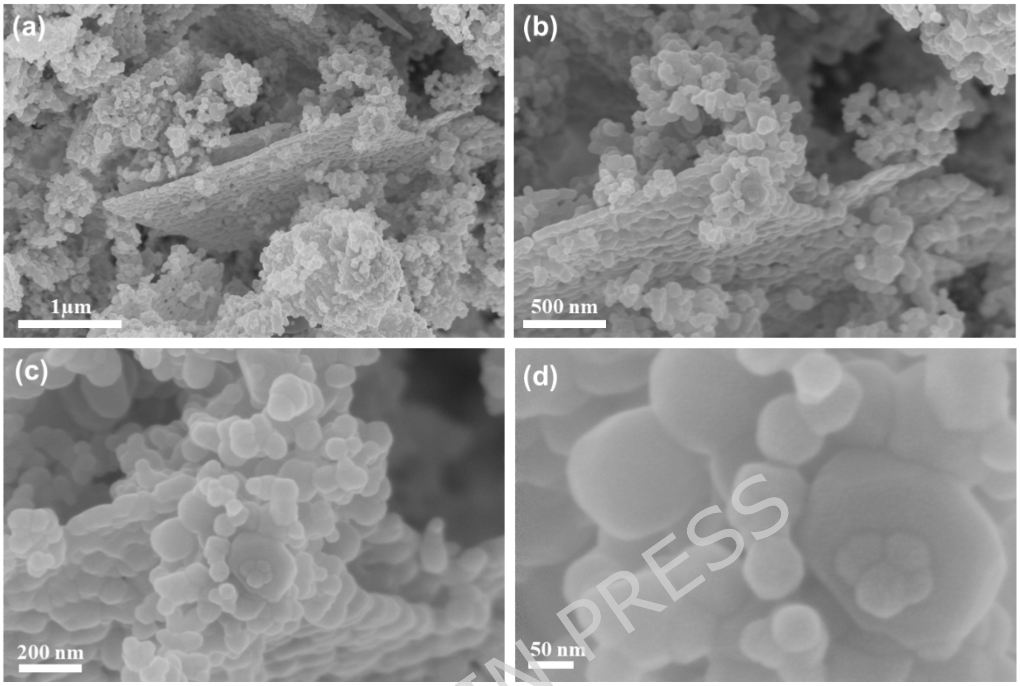

Figure 2. FE-SEM of Ni doped Co_3O_4@G.

Transmission electron microscopy (TEM) analysis provides further insights into the nanostructure and elemental composition of the Ni-doped Co_3O_4@G composite. Low-magnification TEM images (Fig. 3a and 3b) corroborate the spherical morphology of the Ni-doped Co_3O_4 nanoparticles, with uniform sizes ranging from 30 to 60 nm, densely anchored on the semi-transparent graphene nanosheets, maintaining the hierarchical and porous arrangement observed in SEM. High-resolution TEM (Fig. 3c) reveals well-defined lattice fringes with interplanar spacings of approximately 0.24 nm attributable to the (311) planes of the cubic spinel Co_3O_4 phase, respectively, demonstrating high crystallinity and structural integrity despite Ni doping[29].

Additionally, energy-dispersive X-ray spectroscopy (EDS) elemental mapping (Fig. 3d and 3e) highlights the homogeneous distribution of constituent elements: Ni (green), O (blue), Co (magenta), and C (red), where Ni and Co are colocalized within the nanoparticles, while C predominantly delineates the graphene support, evidencing successful Ni incorporation into the Co_3O_4 lattice and effective hybridization with the graphene matrix[30,31]. This detailed nanoscale characterization underscores the material's potential for enhanced electrochemical performance by promoting efficient charge transfer and active site exposure.

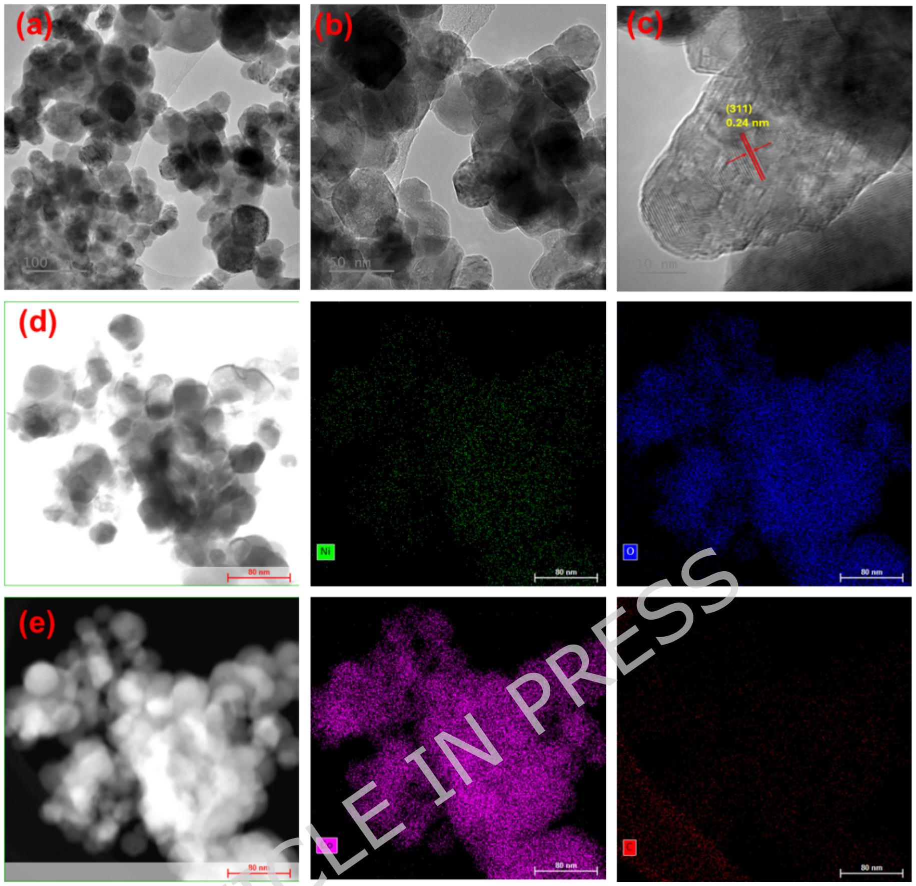

Figure 3. HR-TEM of Ni doped Co_3O_4@G.

The chemical composition and states of Ni-doped Co_3O_4@G composite materials were studied using XPS (Figure 4). The C 1s spectrum typically displays several components corresponding to different carbon functionalities within the graphene backbone. The Figure 4a shows peaks at 284.67 eV (sp^2/sp^3 hybridized C=C/C–C in the graphene backbone), 286.08 eV (C–O), and 288.49 eV (C=O/O–C=O, carbonyl/carboxyl)[32,33]. These results confirm the partial oxidation of graphene, with retained functional groups that improve conductivity and interfacial interactions in the composite[34,35]. The Co 2p spectrum Fig. 4b exhibits spin-orbit doublets with peaks at 779.26 eV/794.37 eV (assigned to Co^{3+}) and 780.84 eV/796.08 eV (assigned to Co^{2+}), accompanied by characteristic shake-up satellites at 789.15 eV and 804.12 eV (primarily from Co^{2+}). This confirms the mixed valence states typical of the spinel Co_3O_4 structure[36]. The relative peak intensities of the lower-binding-energy components suggest a predominant Co^{3+} species, yielding an estimated Co^{3+}/Co^{2+} ratio of approximately 2:1[37] (close to the stoichiometric value in Co_3O_4), with possible slight enrichment in Co^{3+} due to Ni doping.

Such a high Co^{3+} content, along with modulated electronic structure, is known to facilitate optimal adsorption of OER intermediates and improve catalytic activity[38,39]. The Ni 2p spectrum Fig. 4c reveals main peaks at 853.80 eV (Ni^{3+}) and 855.48 eV (Ni^{2+}) for the $2p_{3/2}$ region, with corresponding $2p_{1/2}$ peaks at 871.31 eV and 873.01 eV, and satellite features at 861.06 eV and 879.51 eV. This indicates successful Ni incorporation in mixed Ni^{2+}/Ni^{3+} states within the Co_3O_4 lattice, where Ni^{3+} species likely contribute additional active sites for OER[27,40]. The O 1s spectrum Fig. 4d deconvolutes into three components at 529.50 eV (lattice oxygen in M–O bonds of the spinel structure), 531.54 eV (oxygen atoms at the surface that are partially replaced by hydroxyl groups), and 533.11 eV (adsorbed oxygen species or oxygenated functional groups from graphene)[41–43]. The significant contribution from surficial and/or hydroxyl oxygen indicates abundant surface defects, which serve as active sites for enhanced oxygen evolution reaction (OER) performance[44].

Overall, the XPS analysis verifies the formation of Ni-doped Co_3O_4 supported on partially oxidized graphene, featuring mixed metal valence states, rich oxygen defects, and functional groups. These structural and electronic features synergistically optimize the adsorption/desorption of oxygen intermediates, leading to superior OER electrocatalytic performance.

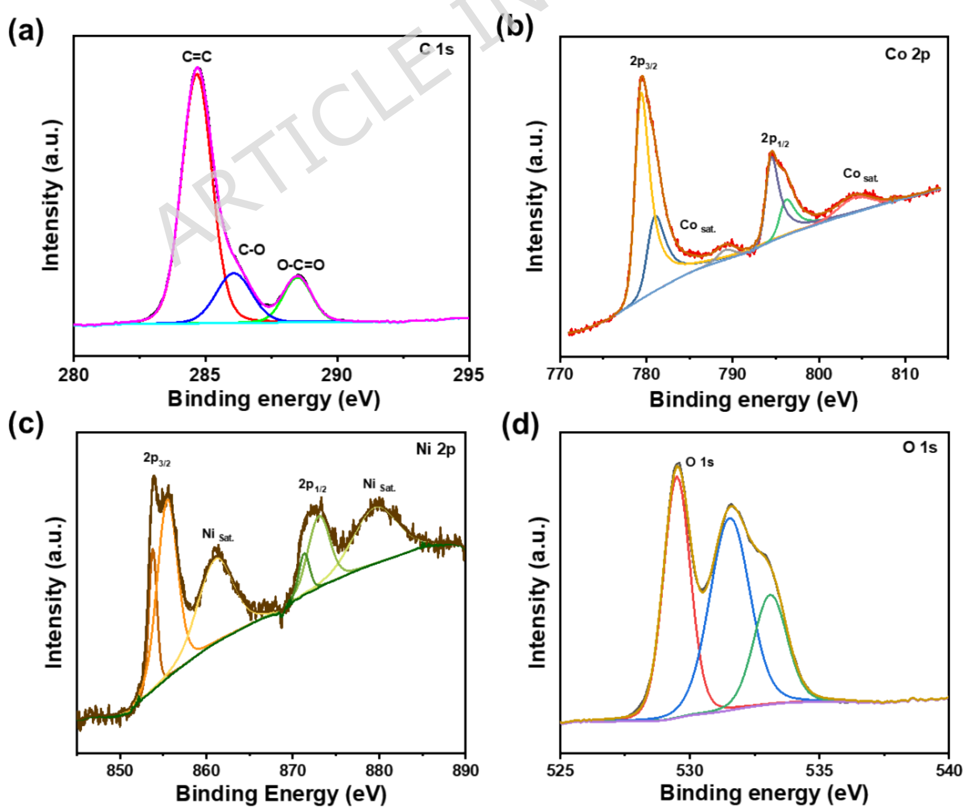

Figure 4. XPS Spectra of Ni doped Co_3O_4@G: (a) C 1s, (b) Co 2p, (c) Ni 2p, (d) O 1s.

3.2 Electrocatalytic Performance for OER

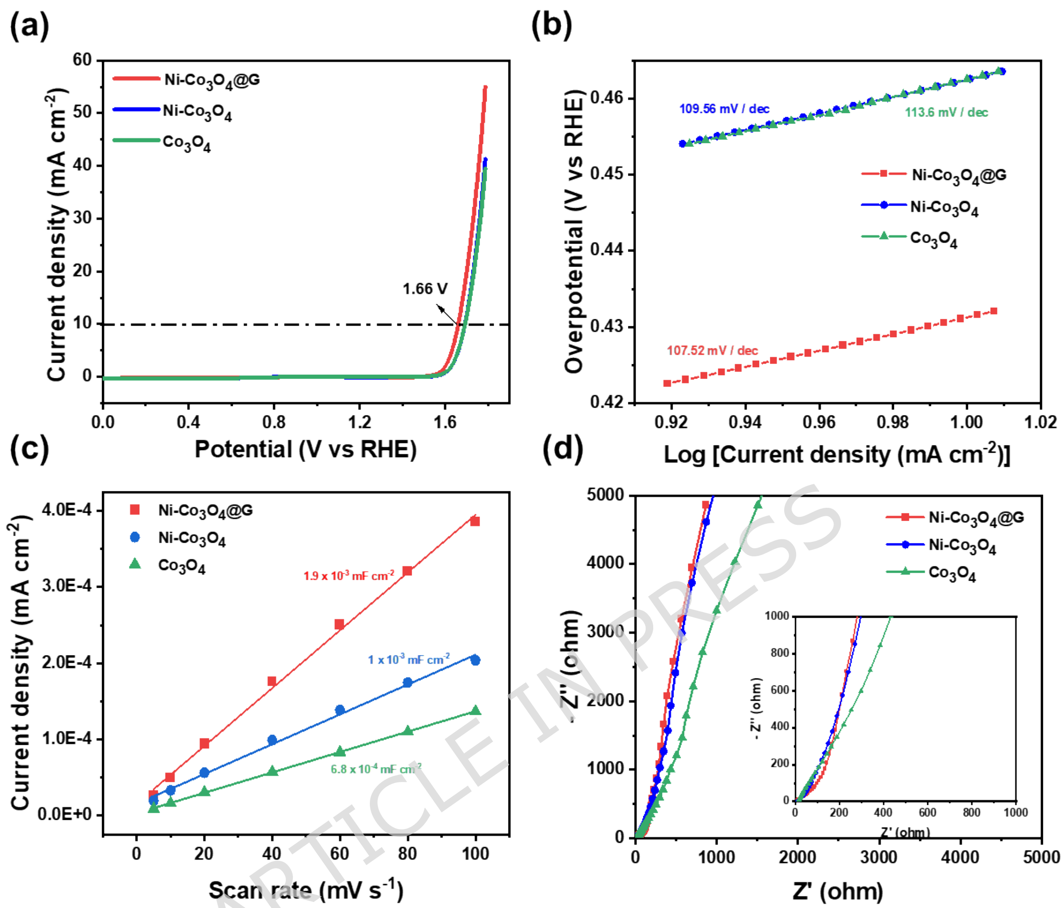

Figure 5. Electrocatalytic OER performance of Co_3O_4, Ni-Co_3O_4, and Ni-Co_3O_4@G in 1 M KOH electrolyte: (a) LSV polarization curves (scan rate 5 mV s^{-1}), (b) corresponding Tafel plots, (c) double-layer capacitance (C_dl) derived from CV scans at different rates, and (d) EIS Nyquist plots.

The electrocatalytic oxygen evolution reaction (OER) performance of the as-prepared Co_3O_4, Ni-Co_3O_4, and Ni-Co_3O_4@G catalysts was systematically evaluated in 1 M KOH electrolyte using linear sweep voltammetry (LSV), Tafel analysis, double-layer capacitance (C_dl) measurements, and electrochemical impedance spectroscopy (EIS), as shown in Figure 5. The LSV polarization curves (Figure 5a) demonstrate that the Ni-Co_3O_4@G composite exhibits the highest OER activity, achieving a significantly higher current density at lower potentials compared to Ni-Co_3O_4 and pristine Co_3O_4. Notably, the Ni-Co_3O_4@G catalyst reaches a potential of 1.66 V vs. RHE at high current densities, corresponding to a low overpotential that markedly outperforms the other samples, reflecting its superior catalytic efficiency. This enhanced OER activity of Ni-doped Co_3O_4 is consistent with previous studies showing improved performance upon Ni incorporation[27,45]. This improvement is attributed to Ni-induced enrichment of high-valence Co^{3+}/Ni^{3+} species and abundant oxygen vacancies, which optimize OER intermediate adsorption and lower the reaction energy barrier. In contrast, both Ni-Co_3O_4 and Co_3O_4 require higher potential to attain comparable current densities, highlighting the beneficial roles of Ni doping and graphene incorporation. The incorporation of graphene-based materials has been shown to enhance the electrochemical performance of various metal oxides due to improved conductivity and active surface area[46,47]. Furthermore, a comparison with catalysts synthesized from commercial cobalt precursors (Figure S2, Supporting Information) reveals nearly identical OER polarization behavior, with similar current densities and overpotential values for both Ni–Co_3O_4@G and Ni–Co_3O_4. This result indicates that the use of recycled cobalt does not compromise catalytic performance, supporting its viability as a sustainable alternative.

The Tafel plots derived from the LSV curves (Figure 5b) further elucidate the reaction kinetics. The Ni-Co_3O_4@G catalyst displays the lowest Tafel slope of 107.52 mV dec^{-1}, indicating the fastest OER kinetics and favorable reaction pathways. This is in line with findings that optimized electronic structures and coordination environments, often achieved through doping or composite formation, can significantly improve OER activity[38]. This is followed by pristine Co_3O_4 with a slope of 109.56 mV dec^{-1}, while Ni-Co_3O_4 shows a slightly higher slope of 113.6 mV dec^{-1}. The reduced Tafel slope for the composite suggests that the integration of graphene effectively lowers the energy barrier for the rate-determining step, likely through enhanced adsorption/desorption of oxygen intermediates (*OH, *O, and *OOH). Graphene's functional groups and high surface area contribute to better charge transfer and increased active sites, which are critical for OER kinetics[48]. To probe the electrochemical active surface area (ECSA), cyclic voltammetry was performed in the non-Faradaic region, and the double-layer capacitance (C_{dl}) was calculated from the linear relationship between current density difference and scan rate (Figure 5c). The Ni- Co_3O_4@G catalyst exhibits the highest C_{dl} value of 19 mF cm^{-2} (derived from a slope of 1.9×10^{-2} mA cm^{-2} (mV $s^{-1})^{-1}$), signifying a substantially larger ECSA and greater exposure of active sites compared to Ni-Co_3O_4 ($C_{dl} \approx 10$ mF cm^{-2}) and Co_3O_4 ($C_{dl} \approx 8$ mF cm^{-2}). This enhancement is attributed to the high surface area and improved dispersion provided by the partially oxidized graphene support, which prevents agglomeration and facilitates electrolyte accessibility. Similar benefits of graphene in enhancing surface area and preventing aggregation have been observed in other metal oxide composites[49–51].

Electrochemical impedance spectroscopy (EIS) Nyquist plots (Figure 5d) reveal the charge-transfer characteristics at the electrode/electrolyte interface. The Ni-Co_3O_4@G catalyst shows the smallest semicircle diameter, corresponding to the lowest charge-transfer resistance (R_ct), followed by Ni-Co_3O_4 and Co_3O_4. The steeper low-frequency tail for the composite further indicates faster mass transport, underscoring the excellent conductivity and efficient electron transfer enabled by the graphene network. This reduction in charge transfer resistance is a well-documented advantage of incorporating conductive carbon materials like graphene or graphene in metal oxide catalysts[48,52].

The long-term electrochemical durability of the Ni-Co_3O_4@G catalyst toward the oxygen evolution reaction (OER) was further investigated through continuous cyclic voltammetry (CV) cycling in 1 M KOH electrolyte (Figure 6). The LSV polarization curves recorded before and after 3000 cycles are presented in Figure 6. Notably, the post-cycling curve closely overlaps with the initial one, exhibiting only a slight positive shift. Specifically, the potential required to achieve current densities (10 mA cm^{-2}) increases marginally from 1.66 V to 1.68 V vs. RHE, corresponding to a negligible overpotential penalty of merely 20 mV. This outstanding stability highlights the robust structural integrity of the Ni-Co_3O_4@G composite, where the graphene support effectively anchors the Ni-doped Co_3O_4 nanoparticles, preventing agglomeration, dissolution, or phase degradation under harsh oxidative conditions during prolonged OER operation. Such exceptional durability, combined with the previously demonstrated high activity, positions the recycled-cobalt-derived Ni-Co_3O_4@G as a promising and sustainable electrocatalyst for practical water splitting applications[38].

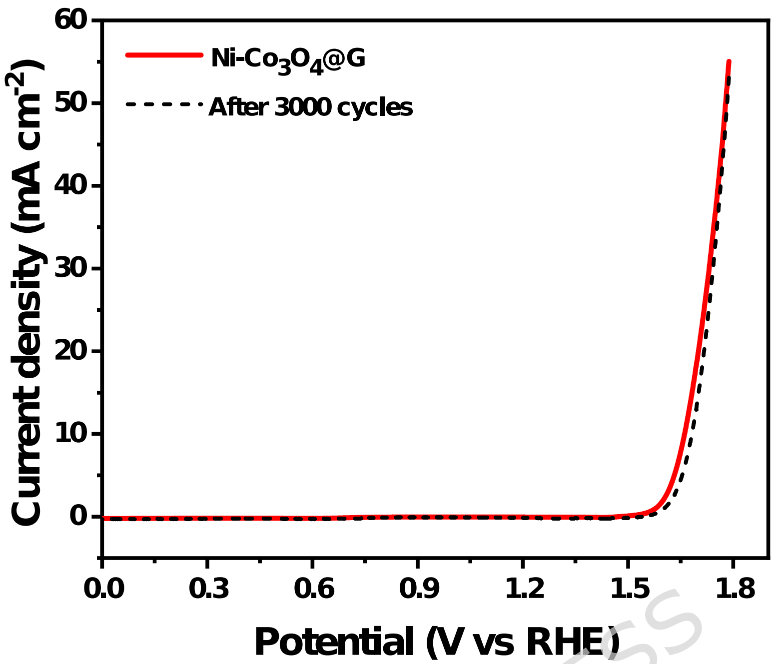

Figure 6. Long-term durability test of the Ni-Co_3O_4@G electrocatalyst for OER in 1 M KOH: LSV polarization curves before and after 3000 continuous CV cycles.

4. Conclusion

In this work, a sustainable strategy was developed to synthesize Ni-doped Co_3O_4@graphene electrocatalysts using cobalt recovered from spent batteries. The successful synthesis of conductive graphene, combined with Ni doping, results in a hierarchical porous architecture with uniformly dispersed nanoparticles, abundant active sites, and significantly enhanced electrical conductivity. Structural analyses confirm the presence of mixed valence states and rich oxygen defects, which are critical for boosting OER activity. Electrochemical measurements demonstrate that the Ni-Co_3O_4@graphene catalyst delivers superior OER performance, achieving a potential of 1.66 V vs. RHE at 10 mA cm^{-2}, a low Tafel slope of 107.52 mV dec^{-1}, high electrochemically active surface area, and low charge-transfer resistance. In addition, the catalyst exhibits excellent durability with only a minor increase of 20 mV after 3000 cycles. Overall, this work presents an effective approach for converting waste-derived cobalt into high-performance electrocatalysts, while highlighting the synergistic role of Ni doping and conductive graphene. The findings provide meaningful insights into sustainable catalyst design for efficient hydrogen production.

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Author contributions

Vu Van Thang, Phuong Nam Le Pham. and Thuy Trang T. Vuong. contribute equally to this work. Vu Van Thang, methodology, formal analysis, investigation, data curation, writing – original draft, writing – review & editing, visualization. Phuong Nam Le Pham and Thuy Trang T. Vuong, formal analysis, investigation, data curation, writing – original draft. Kiem Do Van.: investigation. Tu Le Manh: investigation. Phi Long Nguyen: writing – review & editing, conceptualization, supervision, project administration, funding acquisition.

■ COMPETING INTERESTS

The authors declare that they have no known competing interests.

■ DATA AVAILABILITY

All the data supporting this article has been included in the research article. If the raw data is required, it will be made available on request.

■ FUNDING DECLARATION

This work is financially supported by VinUniversity.

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